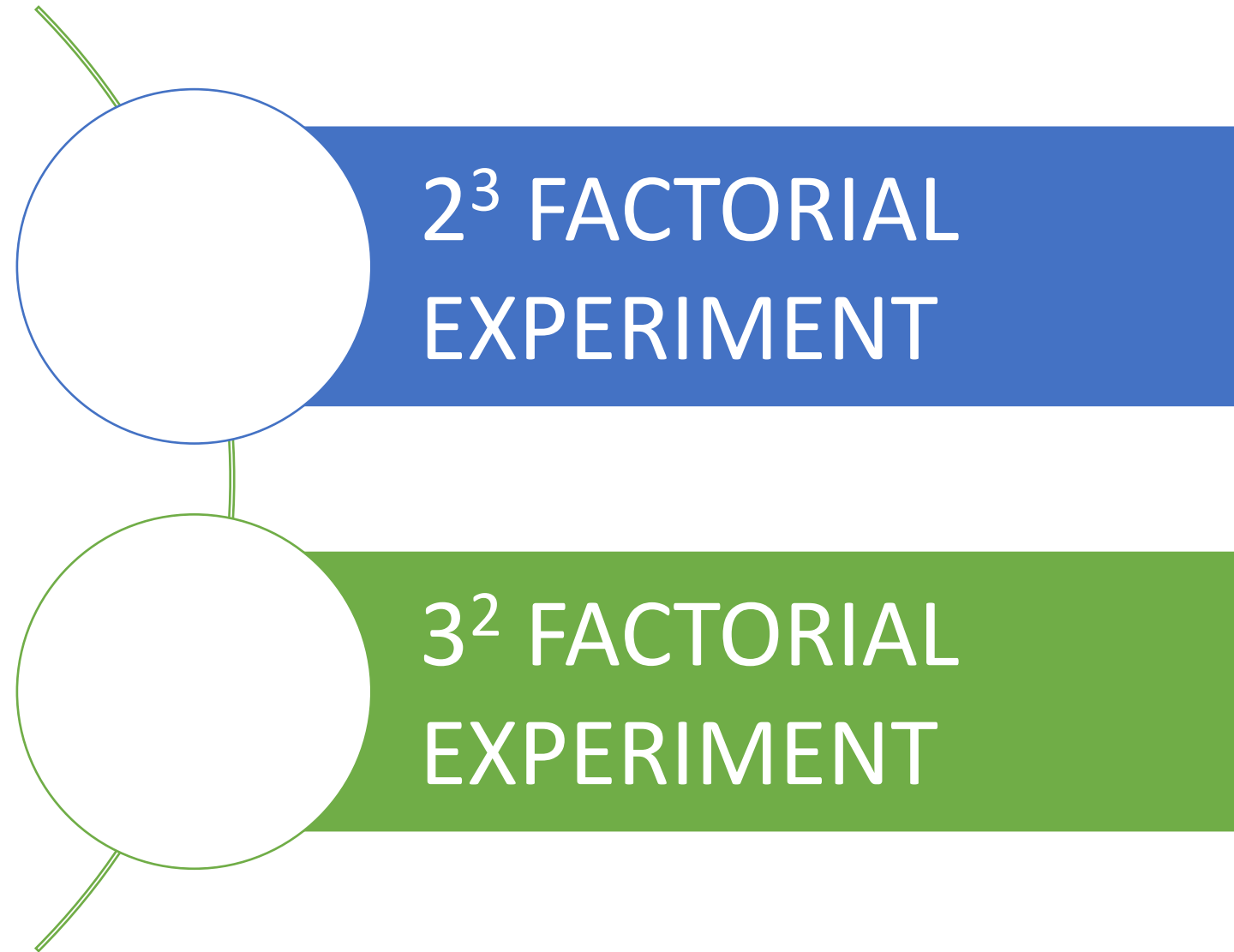




Stat 3113: Lesson 31 Prof Koob's Projects

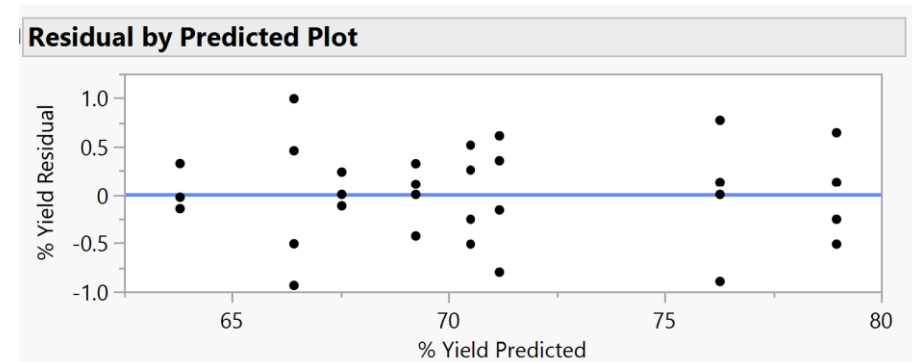
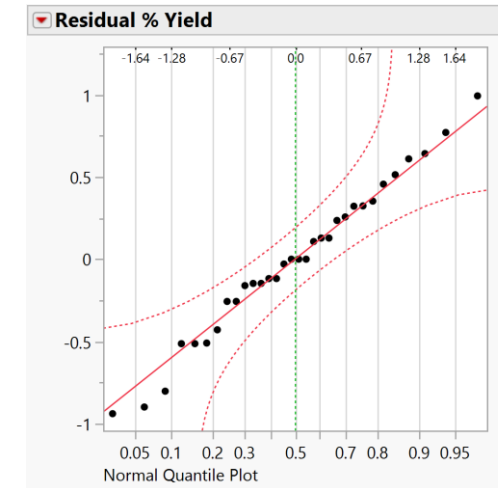
PLAN



Example: 2^3 Factorial Experiment

-
- **Actual Response Variable:** $\% \text{ yield popped} = \frac{\text{weight popped (g)}}{\text{package weight (g)}}$
 - **Experiment Measured Response:** weight popped (g)
 - **Factor 1:** brand
 - **Levels of Factor 1:** Orville Redenbacher, Act II
 - **Factor 2:** butter amount
 - **Levels of Factor 2:** low, high
 - **Factor 3:** time popped
 - **Levels of Factor 3:** 1.5 minutes, 2.5 minutes

Example: Popcorn – check assumptions



- Constant Variance and Normality appear to be met.

Example: Popcorn

Effect Tests					
Source	Nparm	DF	Sum of Squares	F Ratio	Prob > F
Brand	1	1	449.01281	1573.266	<.0001*
Butter	1	1	20.22623	70.8693	<.0001*
Brand*Butter	1	1	2.69529	9.4439	0.0052*
Time (min)	1	1	202.13550	708.2491	<.0001*
Brand*Time (min)	1	1	24.56229	86.0622	<.0001*
Butter*Time (min)	1	1	2.98558	10.4610	0.0035*
Brand*Butter*Time (min)	1	1	9.18993	32.2000	<.0001*

→ STOP because significant.

Step 1: Test for significance of the interaction.

Do we test two-way interaction? Do we test the main effects?

No!

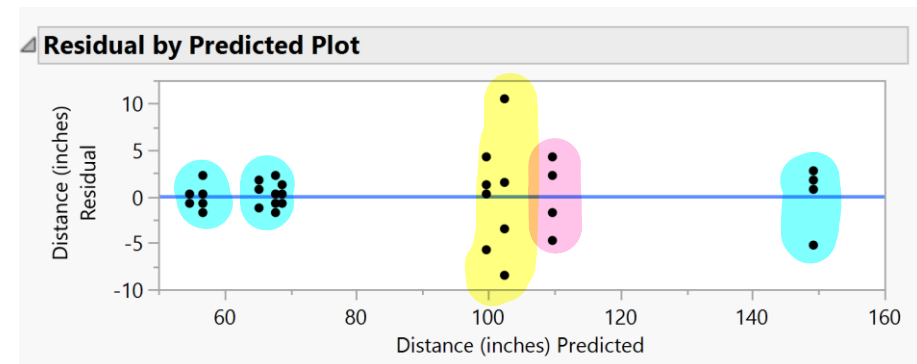
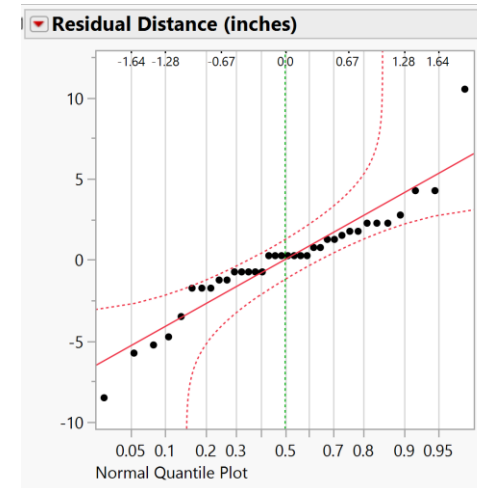
Example: Popcorn

- There is a significant interaction between popcorn brand, butter level, and cooking time on mean %yield (p-value <0.0001).
- Neither two-way interaction or main effects cannot be tested.
- Conclusion: All three factors interact together to impact average %yield.

Example: 3^2 Factorial Experiment

- **Response Variable:** Distance car goes from bottom of ramp
- **Factor 1:** Car type
- **Levels of Factor 1:** sport car, stock car, big truck/bus type
- **Factor 2:** Length of ramp
- **Levels of Factor 2:** high, middle, low

Example: Cars – check assumptions



- Constant Variance is not ok!
- Normality looks fine.

Example: Cars

Effect Tests					
Source	Nparm	DF	Sum of Squares	F Ratio	Prob > F
Car Type	2	2	9718.722	359.4596	<.0001*
Ramp Height	2	2	20026.389	740.7021	<.0001*
Car Type*Ramp Height	4	4	1931.778	35.7247	<.0001*

→ STOP because significant.

Step 1: Test for significance of the interaction.

Do we test the main effects? No!

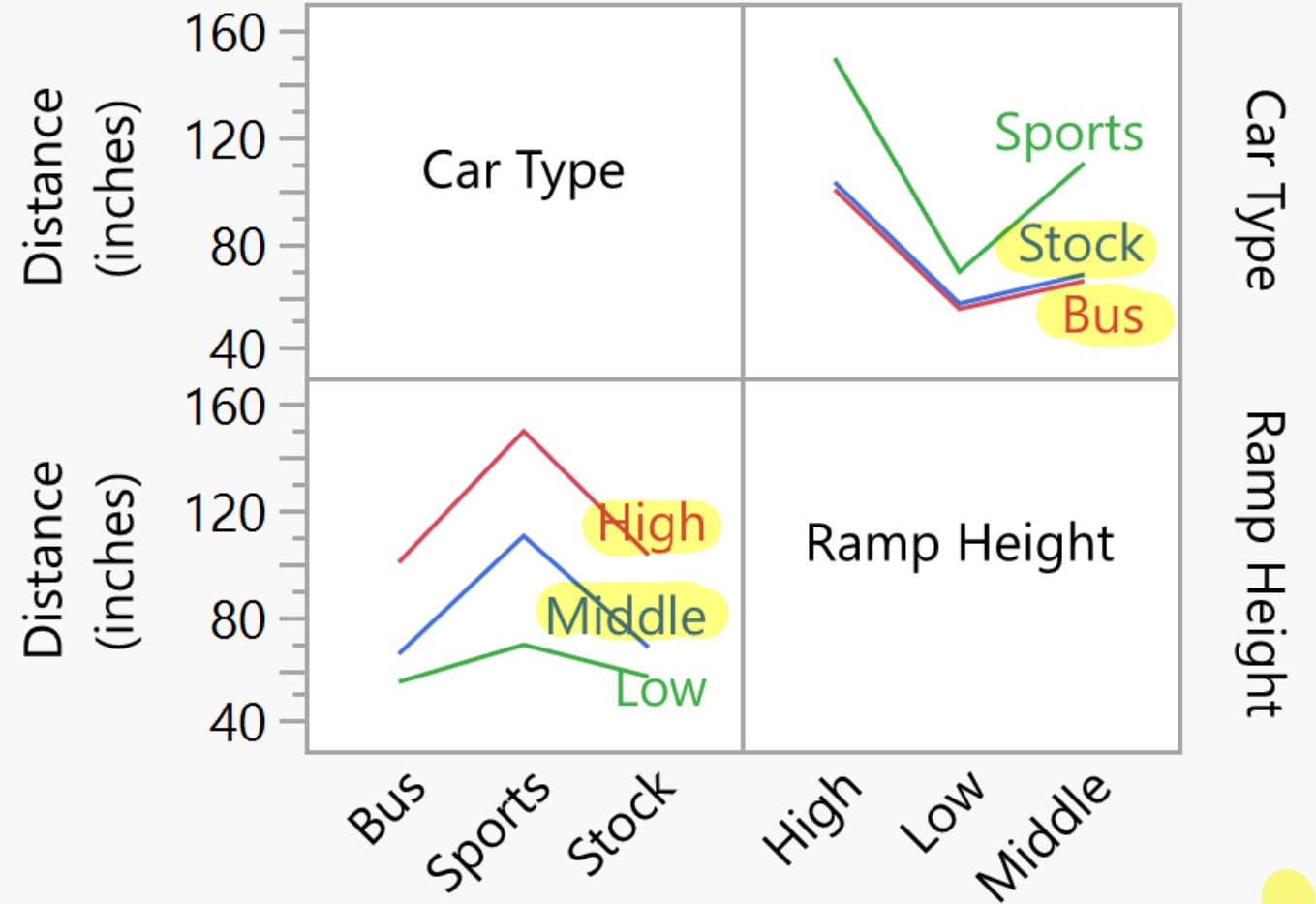


Example: Cars

- There is a significant interaction between ramp length and car type on mean distance traveled ($p\text{-value} < 0.0001$).
- The main effects cannot be tested.
- Conclusion: Both factors interact together to impact average distance traveled.

Example:
Cars

Interaction Profiles





Reading

- CHAPTER 9 – SECTIONS 9.1, 9.2, 9.3, 9.5

Next Topic



Chapter 2 – Section 2.1



Scatterplots - Relationship between two quantitative variables and Correlation